

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – VIVA VOCE (20 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Precision (BL6)	Assessment:	Assessment Tool 3 – Viva Voce
Weightage:	10% of CIA	Total Marks:	20 (20 Questions × 1 Mark)
CO1 Statement:	<i>Create effective and interactive visualizations by integrating data wrangling and advanced visualization tools. (BL6)</i>		
Student Name:	RENEE VINCENT ASR	Reg. No.:	192424161
Section / Batch:	BTECH AI & DS	Date:	14-08-2026

Code of Conduct

I Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renee Vincent

Instructions

- This test contains 20 questions, each carrying 1 mark. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	No. of Questions	Marks
Visualizing Uncertainty	Communicating uncertainty, probabilities, confidence intervals, bootstrapping, hypothetical outcome plots	5	5
Principles of Effective Visualization	Proportional ink principle and axis design	2	2
Handling Overplotting	Overplotting, transparency, jittering	3	3
Common Pitfalls in Visualization	Misuse of color, misleading axes, exaggerating uncertainty	4	4
Accessibility & Uncertainty Design	Accessibility and uncertainty visualization methods	2	2
Advanced Uncertainty Visualization	Probability distributions, combined uncertainty plots, animated/static visualization, interactive techniques	4	4
GRAND TOTAL:			20 Marks

Annexure A – Viva Voce

- 1.Role of visualization in communicating uncertainty in data
- 2.Representation of probabilities effectively in a plot
- 3.Confidence interval and methods to display it visually
4. Bootstrapping and its use in uncertainty visualization
5. Hypothetical outcome plots and situations for their use
6. Principle of proportional ink in data visualization
- 7.Importance of axis design when presenting uncertainty
- 8.Techniques to handle overplotting in scatterplots
- 9.Use of transparency to visualize dense datasets
- 10.Jittering to reduce visual clutter and its limitations
- 11.Common pitfalls in representing uncertainty
- 12.Misuse of color leading to misleading probabilistic plots
- 13.Avoiding misleading axes in visualizations
- 14: Accessibility considerations in uncertainty visualization
- 15: Differences between error bars and gradient plots
- 16: Visualization of probability distributions for categorical variables
- 17: Combining multiple uncertainty representations in one plot
- 18: Pros and cons of animated versus static uncertainty visualizations
- 19: Interactive visualization techniques to understand confidence intervals
- 20: Best practices to avoid exaggerating uncertainty in plots

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (1 Mark)	Level 3 – Proficient (0.75 Mark)	Level 2 – Developing (0.5 Mark)	Level 1 – Beginning (0–0.25 Mark)		Marks
Understanding of Concepts	50%	Demonstrates thorough and accurate understanding of concepts	Shows good understanding with minor errors	Basic understanding with limited clarity	Poor understanding or incorrect responses		
Critical Thinking & Response	50%	Provides logical, confident, and well-structured responses	Gives relevant responses with minor gaps	Responses are partially correct with limited analysis	Unable to analyse or respond effectively		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

MOCK VIVA TEST REPORT

Assessment Tool 3 – Viva Voce

Course Code	DSA06
Course Title	Data Handling and Visualization
Assessment	Assessment Tool 3 – Viva Voce
Total Questions	20
Maximum Marks	20
Marks Obtained	19.75 / 20
Percentage	98.75%
Overall Grade	Excellent

1. Assessment Overview

The mock viva was conducted using the uploaded Assessment Tool 3 – Viva Voce. The source assessment contains 20 questions, each carrying 1 mark, for a total of 20 marks. The covered areas are Visualizing Uncertainty, Principles of Effective Visualization, Handling Overplotting, Common Pitfalls in Visualization, Accessibility & Uncertainty Design, and Advanced Uncertainty Visualization.

2. Overall Performance

Metric	Result
Total Questions	20
Maximum Marks	20
Marks Obtained	19.75
Percentage	98.75%
Full-Mark Responses	19
Partially Correct Responses	1
Incorrect Responses	0
Performance Level	Level 4 – Exemplary

3. Topic-wise Performance

Topic	Questions	Score	Performance
Visualizing Uncertainty	Q1–Q5	4.75 / 5	95%
Principles of Effective Visualization	Q6–Q7	2 / 2	100%
Handling Overplotting	Q8–Q10	3 / 3	100%
Common Pitfalls in Visualization	Q11–Q13	3 / 3	100%
Accessibility & Uncertainty Design	Q14–Q15	2 / 2	100%
Advanced Uncertainty Visualization	Q16–Q20	5 / 5	100%
TOTAL	20	19.75 / 20	98.75%

4. Question-wise Evaluation Summary

Question	Score	Evaluation
Q1	0.75 / 1	Correct core idea; could include variability, probability, or confidence intervals.

Q2–Q20	19 / 19	Accurate, relevant, concise, and well-structured responses.
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5. Strengths

- Strong conceptual understanding of data visualization.
- Excellent knowledge of uncertainty representation.
- Clear and concise viva responses.
- Good understanding of overplotting and methods for reducing visual clutter.
- Strong awareness of accessibility and misleading visualization practices.
- Excellent understanding of advanced uncertainty visualization techniques.
- Able to identify advantages and limitations of visualization methods.

6. Areas for Improvement

The main improvement area is to expand very short answers when the examiner asks for elaboration. For example, the response to Q1 was correct but could have included specific forms of uncertainty such as variability, probabilities, or confidence intervals. For the actual viva, a useful structure is: definition + technique/example + purpose.

7. Rubric-based Assessment

Criterion	Assessment
Understanding of Concepts (50%)	Level 4 – Exemplary
Critical Thinking & Response (50%)	Level 4 – Exemplary
Overall Performance	Level 4 – Exemplary

8. Final Faculty-style Remark

The student demonstrates excellent conceptual understanding and strong response skills in Data Handling and Visualization. The student is able to clearly explain uncertainty visualization techniques, visualization principles, overplotting methods, accessibility considerations, and advanced visualization approaches. Minor improvement is recommended in providing additional examples and elaboration when answering conceptual questions. Overall performance is exemplary.

9. Final Recommendation

Student is well prepared for the Assessment Tool 3 Viva Voce. Recommended revision: Q1 and detailed examples of each visualization technique.

Source basis: Uploaded Assessment Tool 3 – Viva Voce, DSA06 – Data Handling and Visualization.